

# NarraVoice Studio

*Companion Technical Document & Hardware Requirements Addendum*

## Minimum & Recommended System Requirements

Because NarraVoice operates **100% locally and privately** without relying on external cloud APIs, your system hardware directly determines rendering speeds.

| Component        | Minimum Specification                  | Recommended Specification                        |
|------------------|----------------------------------------|--------------------------------------------------|
| Operating System | Windows 10 / 11 (64-bit)               | Windows 11 (64-bit)                              |
| Processor (CPU)  | Intel Core i5 / AMD Ryzen 5 (4+ Cores) | Intel Core i7 / AMD Ryzen 7 (8+ Cores)           |
| Memory (RAM)     | 8 GB RAM                               | 16 GB or higher (Crucial for Qwen model caching) |
| Storage Space    | 5 GB free space (SSD preferred)        | 10 GB free space on a high-speed NVMe SSD        |
| Audio Output     | Windows Compatible Sound Card          | DirectX / ASIO compliant hardware                |

## Known System Limitations & Technical Notes

- **Kokoro Prosody Limits:** Most engine voices sound flat on question intonations. For the best narrative outcome, use *bf\_alice*, *af\_jessica*, or custom voice blends.
- **Pitch Controls:** The pitch slider modifies the global pitch envelope, not the micro-intonation contour lines.
- **ONNX Warmup Friction:** The underlying ONNX runtime requires a brief warmup phase which can cause an audible 'click' at the immediate start of short audio segments.
- **Process Cancellation:** Audio generation cancellation requests are not instantaneous when executing deep ONNX operations.
- **Qwen Engine Prerequisites:** The interactive Qwen architecture requires a functional Python environment, local background server orchestration, and sufficient disk allocation for large model weights.
- **Session Volatility:** Session Instruct profiles do not persist across application restarts by intentional design constraints.

- **Multi-Preset Resampling:** Resampling pitched Kokoro segments down to 24,000 Hz for complex multi-preset voice mixes is theoretically lossy. However, in practice, it is entirely indistinguishable for human speech. *(Engine fix deployed August 2026 to resolve the legacy silent mislabel bug).*

## Underlying Framework & Core Dependencies

NarraVoice builds upon the following robust open-source libraries to ensure high-fidelity execution without cloud tracking:

- **Audio Generation & Engine:** KokoroSharp and KokoroOnnx run the local text-to-speech loops.
- **Playback & Conversion:** NAudio and NAudio.Lame handle master audio playback and high-quality MP3 file export matrices.
- **UI Presentation Layer:** AvalonEdit handles text editing mechanics; ScottPlot drives the voice visualizer data graphics.
- **Orchestration Server:** Python version 3.13.14 is required strictly to host the localized background Qwen server ecosystem.
- **Phoneme Translation:** eSpeak NG acts as the default smart International Phonetic Alphabet (IPA) parsing fallback engine.

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